

# Sumin Jung

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## Research Interests

Large-scale multimodal and foundation models for medical AI; vision-language learning; medical image analysis; clinical representation learning across institutions, modalities, and tasks.

## Education

**Korea University**, Seoul, Republic of Korea Mar 2023 – Feb 2025  
*M.S. in Computer Engineering*

- Advisor: Prof. Jin Tae Kwak; GPA: 4.0/4.5
- **Selected coursework:** Medical Imaging; Efficient AI and Hardware Design; Advanced Digital Signal Processing; Random Signal Analysis

**Dongduk Women's University**, Seoul, Republic of Korea Mar 2019 – Feb 2023  
*B.S. in Computer Science*

- GPA: 3.72/4.5

## Research Experience

**3D-MoReT: Collateral Imaging from 4D MR Perfusion** 2023 – 2024  
*First Author and Lead Researcher, Korea University* [Code](#) [↗](#)

- Led the study from problem formulation and data preprocessing through model design, implementation, experiments, comparative analysis, and initial manuscript drafting.
- Designed a 3D CNN-Transformer regression model combining MobileNetV2, MobileViT, ViT, and residual/skip connections to generate five-phase collateral maps from 4D DSC-MRP.
- Evaluated the model on 952 patients from two university hospitals; outperformed similarly sized models and matched substantially heavier baselines.
- Published the work as first author in the *International Journal of Computer Assisted Radiology and Surgery* and presented it at CARS 2024.

**PathRadX: Pathology Foundation Models for Radiology** 2025  
*Contributing Researcher, Korea University*

- Supported data preparation, result analysis, and manuscript drafting for a study evaluating whether a frozen pathology foundation model could transfer to radiology tasks.
- Contributed to experiments spanning MURA, RSNA pneumonia, and MedFMC datasets and comparisons of modality-adaptation and task-classification strategies; collaborators led model design, implementation, and experiment execution.
- Published in the MedAGI 2025 workshop proceedings held with MICCAI 2025.

**ISLES 2024: Multimodal Infarct Segmentation** 2024  
*Research Design and Analysis, Korea University*

- Contributed problem definition, data preparation, method and experimental design, comparative analysis, and abstract preparation for a 3D Vision Transformer using NCCT, CTP, CTA, and Tmax inputs.
- Collaborated with a teammate who implemented the code and ran the experiments.

## Industry Experience

**DeepClue Inc.** Sep 2024 – Jun 2026  
*Team Leader, Engineering & Manufacturing Team* Seoul

- Led the translation of collateral-imaging research into DeepClue-Precise MRI/CT software as a medical device (SaMD), coordinating clinical requirements, system architecture, implementation, verification, regulatory documentation, and team operations.
- Developed the MRI application and image-processing pipeline end to end—including DICOM preprocessing, image analysis, GUI, output generation, and testing
- Designed the reusable application framework adopted by the CT product and collaborated with the developer responsible for its CT-specific core image-processing pipeline.
- Designed and executed software verification and validation, prepared regulatory and quality-system documents,

coordinated with hospitals and external organizations, and led activities supporting MFDS Class I medical-device registration.

- Supported hospital prototype deployment and converted clinical feedback into software requirements and validation criteria.

## Publications

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- Sumin Jung**, Hyun Yang, Hyun Jeong Kim, Hong Gee Roh, and Jin Tae Kwak. “3D mobile regression vision transformer for collateral imaging in acute ischemic stroke.” *International Journal of Computer Assisted Radiology and Surgery*, 19(10):2043–2054. doi:10.1007/s11548-024-03229-5 [🔗](#). **First author.** 2024
- Hee Jong Ki, Hong Gee Roh, Jin Tae Kwak, In Seong Kim, Jeong Jin Park, Yoo Sung Jeon, Hyun Yang, **Sumin Jung**, Ji Sung Lee, and Hyun Jeong Kim. “The CT collateral map: collateral perfusion estimation and baseline lesion assessment after acute anterior circulation ischemic stroke.” *La Radiologia Medica*, 130(2):235–247. doi:10.1007/s11547-024-01941-5 [🔗](#). *Contribution: data processing and software provision.* 2025
- Heekyung Kim, Hong Gee Roh, Jin Tae Kwak, Hee Jong Ki, In Seong Kim, **Sumin Jung**, Hyun Yang, Jeong Jin Park, Yoo Sung Jeon, and Hyun Jeong Kim. “Simultaneous generation of color-coded arteriography, venography, and dynamic angiography: methodology and clinical applications in stroke.” *La Radiologia Medica*, 130(11):1820–1826. doi:10.1007/s11547-025-02074-z [🔗](#). *Contribution: data processing and software provision.* 2025
- Hyun Yang, **Sumin Jung**, and Jin Tae Kwak. “From pathology to radiology: evaluating the applicability of pathology foundation models.” In *Foundation Models for General Medical AI (MedAGI 2025)*, pp. 34–43. doi:10.1007/978-3-032-07845-2\_4 [🔗](#). *Contribution: data preparation, analysis, and partial manuscript drafting.* 2025

## Presentations & Challenges

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- Sumin Jung**. “3D Mobile Regression Vision Transformer for Collateral Imaging in Acute Ischemic Stroke.” Poster and three-minute oral presentation, CARS 2024. Jun 2024
- Hyun Yang, **Sumin Jung**, and Jin Tae Kwak. “From Pathology to Radiology: Evaluating the Applicability of Pathology Foundation Models.” Poster, MedAGI Workshop at MICCAI 2025. Sep 2025
- Hyun Yang, **Sumin Jung**, and Jin Tae Kwak. “3D Vision Transformer Based Infarct Segmentation.” ISLES 2024 Challenge. 2024

## Honors & Awards

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- **Best Poster Award**, 38th Annual Conference of the Korean Society of Medical Imaging Informatics, 2024
  - **9th Place**, Ischemic Stroke Lesion Segmentation Challenge (ISLES 2024), 2024
  - **Honorable Mention**, Samsung AI Challenge, Image Quality Assessment, 2023
  - **Gold Prize**, Joint Capstone Design Competition, 2022

## Patent Application

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- Hyun Yang, **Sumin Jung**, and Hong Gee Roh. “Machine Learning Method for Generating Hypoperfusion Lesion Segmentation Maps from Cerebral Collateral Blood Flow Images Obtained Using Multi-Phase Dynamic Images, and a Method and System for Predicting Ischemic Penumbra Area Using Generated Hypoperfusion Lesion Segmentation Maps.” *Republic of Korea Patent Application No. 10-2024-0117934*. **Co-inventor.** Aug 2024
- Co-led development of the 3D medical-image framework for generating hypoperfusion lesion segmentation maps using anatomical information.

## Technical Skills

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- Programming:** Python; C; Java; JavaScript; SQL
- Machine Learning:** PyTorch; TensorFlow; scikit-learn; 3D CNNs; U-Net; Transformers; regression; segmentation; classification
- Medical Imaging:** DICOM; pydicom; MR/CT preprocessing; multi-vendor clinical imaging pipelines
- Scientific Computing:** NumPy; pandas; OpenCV; scikit-image; matplotlib; R
- Development Tools:** Git; Docker; Jupyter; PyCharm; Visual Studio Code; PySimpleGUI; PyQt6